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12 December 1977
SUPERSEDING
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24 August 1973

MILITARY SPECIFICATION

EXTRACTORS, TRACK-SPIKE, GASOLINE- OR DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers power-driven spike extractors for pulling spikes from railroad ties.

1.2 Classification. The extractor shall be one of the following types and classes as specified (see 6.2):

Type I - 36-inch (914 mm) gage
Type II - 39-3/8-inch (1000 mm) gage
Type III - 42-inch (1067 mm) gage
Type IV - 56-1/2-inch (1435 mm) gage
Class 1 - Gasoline-Engine-Driven
Class 2 - Diesel-Engine-Driven

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 2230

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
QQ-S-781	- Strapping, Steel, and Seals.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-P-3321	- Painting and Marking: Railway Motive Power and Work Equipment.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-B-26195	- Boxes, Wood-Cleated, Skidded, Load-Bearing Base.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-F-52723	- Filter Element, Hydraulic, Disposable.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits For Army Materiel.

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FEDERAL

FED. TEST METHOD STD. No. 101 - Preservation, Packaging, and Packing Materials: Test Procedures.

(Copies of specifications and standards required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE STANDARDS AND RECOMMENDED PRACTICES

J512 - Automotive Tube Fittings.
J514d - Hydraulic Tube Fittings.
J516 - Hydraulic Hose Fittings.
J517 - Hydraulic Hose.
J534b - Lubrication Fittings.
J833 - Male Physical Dimensions for Construction and Industrial Equipment Design.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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ENVIRONMENTAL PROTECTION AGENCY

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should reference "Code of Federal Regulations 45 CFR Part 1201" and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Description. The track-spike extractors shall be portable, gasoline- or diesel-engine-driven, hydraulic-powered, one-man operated units for pulling spikes from track ties. The frame shall be constructed of steel and shall include a transversing rail for support and positioning of the spike-puller assembly. Each unit shall be mounted on four ductile iron or cast steel single flanged wheels for manual transit over track. The extractor shall be moved off track by use of rubber tires incorporated on the extractor in wheelbarrow fashion. The extractor shall pull at least 12 spikes per minute from both sides of the rails while working in the same direction. The extractor shall pull spikes for a distance of 1/2 mile (805 m) when tested as specified.

3.2 First article (first-produced extractor). The contractor shall furnish a spike extractor for examination and testing within the time frame specified (see 6.2), to prove that his production methods and choice of design detail will produce extractors that comply with the requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.6).

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3.3 Materials. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.10.)

3.3.1 Threaded parts. Screw threads shall conform to Handbook H28. Pipe threads shall conform to the American National Taper Pipe Thread Series.

3.3.2 Welding. Unless otherwise specified (see 6.2), all welding shall be in accordance with the latest applicable codes of the American Welding Society or American Society of Mechanical Engineers.

3.4 Interchangeability. Parts and components shall be interchangeable between spike extractors.

3.5 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, or replacement. Covers or plates which must be removed for component adjustment, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Clearances shall be made to permit the use of receptacles for collecting drainage.

3.6 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.6.1 Human factors. The characteristics of the extractor shall provide for operation by personnel ranging from small man clothed, through the large man clothed, in accordance with SAE J833. The noise level at the operator's station shall not exceed those specified when tested in accordance with MIL-STD-1474, Category C for steady state noise.

3.7 Overall dimensions and weight. The overall dimensions and weight of the extractor in travel position shall be as follows:

Weight with setoff	1100 pounds (499 kg) (maximum)
Length	74 inches (1880 mm) (nominal)
Width	98 inches (2489 mm) (maximum)
Height	48 inches (1219 mm) (nominal) (pneumatic wheels raised)

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3.8 Bearings. All bearings shall be of the antifriction type designed to carry the radial and axial loads. Plain bearings may be used in lightly loaded positions.

3.9 Setoff. The setoff shall be two pneumatic-tired wheels for off track movement of the extractor in wheelbarrow fashion. Not more than two men shall be required to remove the extractor in wheelbarrow fashion off the rails, to the side, and clear of the track. Removal of the extractor from and clear of the track shall be completed in not more than 3 minutes.

3.10 Braking system. The extractor shall be equipped with a hydraulic or mechanical parking brake system that will hold the extractor in a stationary position when on a 5 percent grade. Wheels shall roll freely when the brake is released.

3.11 Hydraulic system. The hydraulic system shall provide a means of operating the extractor. Oil for the system shall conform to MIL-L-2104, Grade 10. For low temperature operation, oil conforming to MIL-L-46167 may be used. All hydraulic component materials shall be compatible with these oils. Unless otherwise specified herein, all system components shall withstand a proof pressure that is equal or greater than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) without external leakage, damage, or permanent deformation. The hydraulic system shall be cleaned free of foreign matter (including weld slag and spatter). Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly on the vehicle.

3.11.1 Test points. One test point shall be located at the outlet of each pump for measuring pressure.

3.11.2 Cylinders. The extractors shall be provided with two, single or double acting hydraulic cylinders (RAMS). The cylinders shall be fabricated from seamless steel tubing, or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device at the rod to prevent foreign material and fluids from entering and damaging the seal and other cylinder components. All cylinders shall perform as specified herein without buckling or bending. The cylinders shall be located as close as practicable to the track wheels and shall operate from hydraulic pressure supplied by pump(s) powered by the engine. The piston rod shall be hard chrome plated to prevent corrosion. The bearing surface shall be a material that will not scar, score, or gall the inside surface of the cylinder.

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3.11.3 Pumps. The extractor shall be equipped with hydraulic pump(s) driven directly by a gasoline- or diesel-engine. The pump(s) shall be a constant displacement type and shall be capable of developing the system relief valve setting pressure at a speed corresponding to the manufacturer's rated speed.

3.11.4 Controls. Cylinders shall be provided with individual controls. All hydraulic control valves shall be of the metering type so that accurate control of the unit may be maintained at all times. All controls shall be located within easy reach of the operator when he is in his normal operating position. Pushing of a lever shall result in a lowering of the shoe, and pulling on shall result in a raising or retracting the shoe. Forces necessary to actuate the controls shall be 10 pounds (5 kg) maximum. The travel of the controls shall not exceed 6 inches (152 mm) from center position when measured at the end of the grip. The travel control grip shall have different configuration from that of 1-cylinder control levers. All control-valve spools shall be spring centered with sufficient force to return the levers to the center position and retain it there throughout all operations. Valve spools exposed to dirt and moisture shall be provided with wipers and seals. Spools shall not bind under all operating conditions specified herein. Hydraulic control valves shall unload the pump when cylinders or propelling motor are not in use.

3.11.5 Relief valve. A relief valve shall be provided to limit the pump pressure to the maximum required to perform the necessary system functions. Pressure equal to the relief valve setting shall be obtained only when a cylinder is bottomed or when the system is subjected to shock loading.

3.11.6 Return line filter. The system filter element(s) shall conform to MIL-F-52723. The filter shall maintain a full flow condition during all operations of the extractor. The filter bypass may open for system oil at a temperature less than 70° F (21° C). All return oil shall pass through the filter except relief oil which may return directly to the reservoir. The filter shall be located so that the element may be removed for examination or replacement without spilling or drawing oil from reservoir. An automatic bypass that permits full flow at a pressure of not less than 35 psig (241 kPa) shall be provided in the filter housing. The filter shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall be clearly readable without the removal of any components. The indicator shall show when the element is clean and, as a minimum, when the filter is beginning to bypass, requiring a change of the element.

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3.11.7 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibrations that may result in leaks, loosening of fasteners, or cracks. Connections on the reservoir shall be located so that connecting lines can be removed without removing the reservoir or any other major component. The reservoir shall be located and constructed so that during operation and performance of all functions specified herein the pump shall not be subject to aeration. All control linkages or connections into the reservoir shall be sealed to prevent leakage. The reservoir should be baffled when necessary to provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. A dipstick or sight gage shall be furnished to determine the reservoir oil level. The dipstick shall be marked "Add" and "Full" by etching, stamping, engraving, or embossing. The dipstick shall be removable without tools and shall be located in the filler neck. When practical, the sight gage or window should be visible to the operator when he is in his normal position and shall be marked "Add" and "Full" and should be recessed in the reservoir or protected to prevent breakage. The "Add" and "Full" marks on the sight gage window, or dipstick shall indicate the correct oil level when all hydraulic lines and components are full. Access to the interior of the reservoir shall be provided for repair and removal of all internal parts and components. The filler cap and tube shall be located to provide for filling from a standard 5-gallon (19 L) container and to prevent entrance of contaminants throughout filling and checking operations. The reservoir pump suction openings should be located a minimum of 1 inch (25 mm) above the bottom of the reservoir to prevent entrance of settled contaminants. Except for sealed reservoir, a breather shall be provided and shall filter incoming air. The breather shall preclude the entrance of water induced by rain.

3.11.8 Hydraulic lines. Hydraulic lines shall consist of tubing or pipe, hose and fittings. Hose or a combination of hose and tubing shall be used between components that move in relation to each other. All lines shall be routed to provide the shortest line, minimum number of bends, ease of maintenance, and maximum protection. Provision shall be made to prevent damage of lines due to chafing. The hose installation, such as angle of fittings, and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines other than suction lines shall be considered pressure lines. All lines shall be cleaned free of foreign matter including weld slag and spatter.

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3.11.8.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 and pressure hose couplings shall be as specified in SAE J516. The hose selected shall have a working pressure which is equal to or greater than the hydraulic system maximum working pressure (relief valve setting). The hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

3.11.8.2 Pressure tubing and piping. Pressure tubing and piping shall have a minimum burst pressure rating of 3.5 times the system maximum working pressure relief valve setting. All tubing bends shall be smooth without flattening, kinking, or wrinkling of the tube. The term "without flattening" is defined as follows: Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum diameter minus the tube minimum diameter shall be not greater than 15 percent of the tube nominal diameter. All tubing and piping shall be cleaned free of foreign matter including cuttings and weld slag and spatter.

3.11.8.3 Pressure pipe fitting and pressure tube fittings. Unless otherwise specified herein, pressure tube fitting shall conform to either SAE J512 or SAE J514. Pressure tube fitting shall have burst pressure rating that is equal to or greater than 3.5 times the maximum system working pressure (relief valve setting).

3.12 Spike-puller assembly. The extractor shall be equipped with a spike-puller assembly, connected to the hydraulic pump with a flexible hose. The spike-puller assembly shall be positioned on the extractor so that the tongs can be held 2 degrees off the vertical when pulling spikes.

3.12.1 Jaws. The jaws shall be of alloy steel, hardened and tempered, showing a hardness of not less than 43 or not more than 50 on a Rockwell C scale. The jaws shall have sufficient material for regrinding. No more than 15 minutes shall be required to replace the jaws by an experienced operator/mechanic using common handtools.

3.13 Engine. For the purpose of selecting the engine the extractor shall be classified as a Class I end item in accordance with MIL-STD-1410. The engine shall be gasoline or diesel and shall conform to MIL-STD-1410, Class I.

3.13.1 Fuel tank. The capacity of the tank shall be sufficient for 8 hours of continuous operation in operating conditions specified herein. Fuel capacity shall be based upon 85 percent of the fuel consumption rate based on the engine manufactures certified curve at the applicable horsepower and governed speed.

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3.13.2 Accessories. The engine shall be furnished with a housing, and the following instruments, as applicable. When a diesel engine is used an electrical cranking system, battery and battery charging system shall be furnished.

(a) Gage, lubricating oil.
(b) Indicator, fuel, liquid quantity.
(c) Tachometer.
(d) Hourmeter.

3.13.3 Exhaust system. The engine shall be equipped with a muffler and an exhaust pipe located and installed in such a manner that exhaust gases will not be discharged under the extractor or be hazardous to operating personnel.

3.13.4 Crankcase ventilation system. A positive crankcase ventilation system in accordance with EPA requirements shall be provided on each engine, the system shall prevent or eliminate emission of crankcase vapors into the atmosphere under all operational conditions specified herein. If the ventilation system is connected to the air cleaner, it shall be connected in such a manner that no vapors or condensation will come in contact with, or be deposited on the air cleaner element. The design of the system shall be such that service or replacement of the system, or any part thereof, may be accomplished without removal of other components of the engine.

3.14 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.14.1 Lubricants. The extractor shall operate as specified herein when lubricated with military lubricants, or as recommended by the manufacturer.

3.14.2 Lubrication fittings. Lubrication fittings, when used, shall conform to SAE J534b. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun equipped with a hydraulic coupler and a 10-inch (254 mm) flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

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3.14.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.14.4 Enclosure with integral reservoir of lubricant. Enclosures such as gear cases and transmission housing that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/8-inch (3 mm) NPT, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for complete draining. The drain outlet shall be fitted with a magnetic drain plug not less than 1/4-inch (6 mm) NPT. Drainage shall be to the ground when the extractor is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground.

3.14.5 Grease lubrication. All grease lubrication including lubrication of sealed bearings, shall be with grease as recommended by the manufacturer or as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F (121° C).
(b) MIL-G-10924 - For bearings where operating temperatures do not exceed 175° F (79° C).

Each extractor shall be run-in (broken-in) (see 4.5.2) and delivered with these greases. The bearings shall be cleaned before lubrication because some greases are not compatible with others. The extractor shall be tagged in a conspicuous place to indicate which grease has been used.

3.14.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with winter grades of military or commercial oils designated for use in the temperature range of 0° F to 32° F (-18° C to 0° C). The extractor shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.15 Lifting attachments. Permanently affixed attachments that enable the extractor to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the extractor 2-1/2 times the resulting force imposed when the extractor is suspended by the lifting attachments. Attachments for multiple slings shall be located so that:

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(a) The slings will converge not more than 19 feet (6 m) above the lowest extremity of the extractor.
(b) Not less than 1-inch (25 mm) clearance will be maintained between the slings and extractor.

The inside diameter of the attachment eye shall be not less than 3 inches (76 mm). Lifting attachments may also be used as tiedown attachments (see 3.16).

3.16 Tiedown attachments. Permanently affixed attachments that permit fastening the extractor to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand without damage to the extractor, its proportion of the following static loads: 8-G forward, 8-G rearward, 2-1/4-G upward, 1-1/2-G sideward, where G equals cargo weight of extractor. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments (see 3.15).

3.17 Identification marking. The extractor shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514, Type I, Style 2, photographic plate. Transportation-data plates shall conform to MIL-P-514, Type I, Style 4, photographic plate, and shall indicate location and capacity of lifting and tiedown attachments.

3.18 Instruction plates. Each extractor shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedure to be followed in operating or servicing the extractor. Instruction plates shall conform to MIL-P-514, Type I, Style 3, photographic plate.

3.19 Treatment and painting. The portions of the extractor and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-P-3321.

3.20 Toolbox. The extractor shall be provided with a weatherproof toolbox made of steel not less than 0.071 inch thick (2 mm). The toolbox lid shall be secured to one edge of a side by a continuous hinge extending the full length of the toolbox edge. The toolbox lid shall open to at least 90 degrees from the closed position. The lid shall remain in the fully open position either by its own weight or by holding devices. Two spring-tensioned hook-type fasteners shall be provided to keep the lid closed. Not less than a 3/4-inch (19 mm) clearance shall be provided between the hook lever and any part of the toolbox. The fasteners shall be attached to the toolbox by either bolts or rivets. A hasp which will accomodate a key-operated lock shall be provided for locking the toolbox. The toolbox shall be fastened

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o the extractor in a protected position. A drain hole shall be provided e bottom of the toolbox. The toolbox shall be of a size to hold the p ting and maintenance tools and repair parts specified.

3.21 Workmanship.

3.21.1 Steel and aluminum fabrication. Steel and aluminum used in the abrication of the extractors shall be free from kinks and sharp bends. he straightening of material shall be done by methods that will not cause njury to the metal. Shearing and chipping shall be done neatly and accu- ately. Corners shall be square and true. Flame cutting may be employed nstead of sawing or shearing. All bends of a major character shall be made ith metal dies or fixtures to insure uniformity of size and shape.

3.21.2 Bolted connections. Boltholes shall be punched or drilled and hall have the burrs removed. Washers shall be provided in accordance with ommercial practice and all bolts, nuts, and screws shall be tight.

3.21.3 Riveted connections. Rivet holes shall be punched or drilled nd shall have the burrs removed. Rivets shall completely fill the holes. ivet heads shall be full, neatly made, concentric with the rivet holes, nd in full contact with the surface of the member.

3.21.4 Welding. The surfaces of parts to be welded shall be free from ust, scale, paint, grease, and other foreign matter. Welds shall transmit tress without permanent deformation or failure when the parts connected by he welds are subjected to proof and service loading.

QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the ontract, the contractor is responsible for the performance of all inspec- ion requirements as specified herein. Except as otherwise specified in he contract, the contractor may use his own or any other facilities suit- ble for the performance of the inspection requirements specified herein, nless disapproved by the Government. The Government reserves the right to erform any of the inspections set forth in the specification where such nspections are deemed necessary to assure supplies and services conform to rescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible or insuring that components and materials used are manufactured, examined, nd tested in accordance with referenced specifications and standards.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced extractor(s) inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced extractor(s) inspection.

4.3.1 Examination. The first-produced extractor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced extractor shall be tested as specified in 4.5.2 through 4.5.2.9. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each extractor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each extractor shall be run-in for at least 10 minutes before being tested as specified in 4.5.2.8. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The extractor shall be examined for the following defects:

101. Materials not as specified.
102. Threaded parts not as specified.
103. Welding not as specified.
104. Safety not as specified.
105. Human factors not as specified.
106. Overall dimensions and weight not as specified.
107. Bearings not as specified.
108. Setoff not as specified.
109. Braking system not as specified.
110. Hydraulic system not as specified.
111. Test points for measuring pressure not as specified.
112. Cylinders not as specified.
113. Pumps not as specified.
114. Controls not as specified.

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115. Relief valve not as specified.
116. Return line filter not as specified.
117. Reservoir not as specified.
118. Hydraulic lines not as specified.
119. Pressure hose assemblies not as specified.
120. Pressure tubing and piping not as specified.
121. Pressure piping pipe fittings and pressure tube fittings not as specified.
122. Spike-puller assembly not as specified.
123. Jaws not as specified.
124. Engine not as specified.
125. Fuel tank not as specified.
126. Accessories for the equipment not furnished as specified.
127. Exhaust system not as specified.
128. Crankcase ventilation system not as specified.
129. Lubrication not as specified.
130. Lubricants not as specified.
131. Lubrication fittings not as specified.
132. Pressure-release device not as specified.
133. Enclosure that is integral with reservoir of lubricant not as specified.
134. Grease lubrication not as specified.
135. Oil lubrication not as specified.
136. Lifting attachments not as specified.
137. Tiedown attachments not as specified.
138. Instruction plates not as specified.
139. Steel and aluminum fabrication not as specified.
140. Bolted connections not as specified.
141. Riveted connections not as specified.
142. Welding not as specified.
143. Identification marking not as specified.
144. Treatment and painting not as specified.
145. Toolbox not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the extractor shall be lubricated withoils and greases specified in 3.14. Oils shall be those designated for use in the ambient temperature at the place of test. The test specified in 4.5.2.4 shall be executed by one man. During all tests, the engine shall be governed at a speed not greater than the continuous rated speed as established by the qualification test. The hydraulic pump shall not exceed the continuous rpm recommended by the pump manufacturer when operating at not more than 3,000 psi (21 MPa) during the tests. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test.

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4.5.2.2 Hydraulic system. Pressurize the hydraulic system to not less than 1-1/2 times the normal operating pressure required to meet the requirements of this specification. Any external leaks, deformation, or breakage of any component shall constitute failure of this test.

4.5.2.3 Spike pulling. Pull spikes from ties with the extractor at a rate of not less than 12 spikes per minute for a distance of 1/2 mile (800 m) with 100 yards (91 m) being on a 5 percent grade. Then pull 16 spikes, four from each side of each rail. Inability to pull spikes from ties at the rate of 12 per minute on the specified gage track, or from both sides of the rails, or failure of the brakes to hold the extractor stationary on a 5 percent grade shall constitute failure of this test.

4.5.2.4 Jaw replacement. Replace the jaws in the spike puller assembly using common handtools. Inability to replace the jaws in 15 minutes using common handtools shall constitute failure of this test.

4.5.2.5 Setoff and off-track movement. Place the extractor in the spike-pulling position with the pneumatic tires locked in the up position at the start of the test. Use two men to set the extractor off and clear of the track. While in wheelbarrow position, move the extractor a distance of 50 feet (15 m) over ballast by rolling on the pneumatic tires. Inability of the two men to complete the setoff portion of the test in 3 minutes or to move the extractor the required distance shall constitute failure of this test.

4.5.2.6 Rail transit. Move the extractor over 100 yards (91 m) of level track on four wheels by one man. Inability to transport the extractor the required distance manually by not more than one man shall constitute failure of this test.

4.5.2.7 Lifting and tiedown attachments. The force specified in 3.15 for lifting attachments and the static load specified in 3.16 for tiedown attachments shall be applied to the attachment when mounted on its structural member. Weld failure or permanent deformation of the attachment, the fastening, or the structural member shall constitute failure of this test.

4.5.2.8 Operational test. The extractor shall be operated for not less than 15 minutes performing all operational functions under specified load to determine performance of all parts. Any operational difficulties, breakage, evidence of contamination on hydraulic filter, leakage, or permanent deformation shall constitute failure of this test.

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4.5.2.9 Noise level measurement. Measure the noise at the operator's station in accordance with MIL-STD-1474 for steady state noise while operating the extractor. Nonconformance to 3.6.1 shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. The preproduction pack of the boxed extractor for Level A or B shall be subjected to the pendulum-impact test of FED. TEST METHOD STD. No. 101, Method 5012, except the vertical height of drop shall be 18 inches (457 mm). Any shifting of contents, loosening or breaking of holddowns, ties, straps, blocking, or bracing or any visible damage to the contents shall constitute failure of the test and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Inspection stages. Inspection shall be in two stages as follows:

(a) The first stage shall include inspection of preservation, and unit marking.
(b) The second stage shall include inspection of materials and marking after closing and strapping of the container.

4.6.2.1.1 Unit of product. For the purpose of inspection, a unit of product for the first stage of inspection shall be a complete pack with the box top not in place. For the second stage of inspection a unit of product shall be a completed pack prepared for shipment.

4.6.2.2 Examination. Each extractor shall be examined for the following defects:

146. Preproduction pack not as specified.
147. Preservation not Level A or Commercial as specified (see 6.2).
148. Preservatives for Level A not as specified.
149. Unprotected surfaces not preserved for Level A as specified.
150. Engine not preserved for Level A as specified.
151. Hydraulic system not preserved for Level A as specified.

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152. Exposed and enclosed gears not preserved for Level A as specified.
153. Preservation of the extractor for Commercial level not as specified.
154. Packing not Level A, Level B, or Commercial as specified.
155. Each extractor not packed for Level A as specified.
156. Each extractor not packed for Level B as specified.
157. Each extractor not packed for Commercial level as specified.
158. Tools not preserved for Level A as specified.
159. Repair parts not preserved for Level A as specified.
160. Marking for military levels of protection not in accordance with MIL-STD-129.
161. Marking for Commercial level of packaging not as specified.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the preparation for delivery requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.7). The preproduction pack may be accomplished utilizing either the preproduction extractor or a production extractor. If the preproduction model extractor is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the preproduction model extractor and the production extractor.

5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.2 Unprotected surfaces. Unpainted exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

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5.2.1.3 Engine. The engine and engine accessories shall be preserved in [illegible]dance with the Level A requirements of MIL-E-10062, as specified for [illegible]II, Method 1.

5.2.1.4 Hydraulic system. The hydraulic fluid supply tank shall be filled o the operating level with the hydraulic fluid recommended for operation. he pistons shall be fully retracted into the cylinder and secured, with any npainted portion of the piston rods (ramshafts) coated with Type P-1 pre- ervative. Where pistons cannot be fully retracted, exposed areas of normally etractable surfaces shall be coated with Type P-6 or P-11 preservative and rapped or covered with barrier material conforming to MIL-B-121, Type I, rade A, Class 2, extending the wrap approximately 2 inches (50 mm) onto the djacent exterior ram cylinder. The wrap shall be secured in place with ape conforming to PPP-T-60, Type IV. The hydraulic control valves shall e secured in the neutral position and preserved in the same manner as pecified for piston rods. Hoses shall not be disconnected. A tag conform- ng to UU-T-81, Type A, shall be attached in a conspicuous location. Infor- ation on the tag shall be applied with a waterproof ink and shall indicate: The hydraulic supply tank is filled with the fluid required for operation."

5.2.1.5 Gears.

5.2.1.5.1 Exposed gears. After the exposed gears have been cleaned and ried, they shall be coated with Type P-1 preservative or with primer con- orming to TT-P-664.

5.2.1.5.2 Enclosed gears. Gear housings shall be filled to the operating evel with the operational lubricant and the gears actuated to assure coating [illegible] interior parts and surfaces with the lubricant. A tag conforming to U-T-81, Type A, shall be attached in a conspicuous location. Information n the tag shall be applied with a waterproof ink and shall indicate: "The ear housing is filled with the approved lubricant. Do not drain."

5.2.1.6 Tools. Maintenance and operating tools shall be preserved in ccordance with the Level A requirements of PPP-P-40.

5.2.1.7 Repair parts. The preservative application criteria and appli- able method(s) of preservation contained in MIL-P-116 shall be used to reserve the repair parts.

5.2.1.8 Consolidation. Components not attached to the extractor shall e consolidated together in the toolbox as space permits. When sufficient pace is not available in the toolbox the additional components shall be onsolidated together in a close-fitting box conforming to PPP-B-601, Over- eas Type, Grade B, style optional. The contents shall be blocked and raced to prevent movement or damage. Strapping of the box shall not be equired.

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5.2.2 Commercial. Each complete extractor and all items issued with the extractor shall be preserved to afford adequate protection against mechanical and physical damage. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings. Items requiring protection from physical damage shall be protected by wrapping or cartonizing of the individual item. All lubrication reservoirs, sumps, etc., shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which military lubricant has been used, the temperature range of lubricant used and grade of lubricant used. A deprocessing guide shall be furnished indicating in detail the preservation materials to be removed and any other servicing required prior to placing the extractor in operation.

5.3 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.3.1 Level A. Each complete extractor, preserved as specified in 5.2, shall be packed in a box conforming to MIL-B-26195, Type II, Style C, Class 1, with rubbing strips and cleated-plywood superstructure. Blocking, bracing, and anchoring of the contents shall be in accordance with MIL-STD-1186. Strapping for securing of the box shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.3.2 Level B. Packing shall be as specified in 5.3.1 for Level A except the box shall be Type I and strapping shall be Finish A.

5.3.3 Commercial. Each complete extractor preserved as specified in 5.2, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for commercial packaging shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

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5.5 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The extractor is intended for use in pulling spikes from railroad ties in preparation for rail laying and retimbering operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of extractor required (see 1.2).
(c) Time frame required for submission of first-produced extractor (see 3.2).
(d) If welding is to be other than specified in applicable codes of AWS (see 3.3.2).
(e) Time frame required for submission of preproduction pack (see 5.1).
(f) Degree of preservation and packing required (see 5.2 and 5.3).
(g) When other than Finish B strapping is required (see 5.3.1).

6.3 Type IV. Type IV extractors are no longer required to be dual gage (56-1/2-inch and 60-inch gages) (1435 mm and 1524 mm gages).

6.4 Lubricants. When requested by the contractor, the contracting officer should obtain a list of military lubricants applicable to the extractor covered by this specification.

6.5 Government serial numbers. The contracting officer should furnish the Government serial numbers for the extractors upon award of the contract or order.

6.6 First-produced extractors. Any changes or deviations of production extractors from the approved first-produced extractor during production will be subject to the approval of the contracting officer. Approval of the first-produced extractor will not relieve the contractor of his obligation to furnish extractors conforming to this specification.

6.7 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve pack, and mark the extractors in accordance with this specification.

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6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, and contractor's maintenance and operational manual to be furnished with each extractor.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary including any special tools and instructions on shipment of extractors. A suggested paragraph is as follows:

"Shipment of extractors shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.10 Recycled material. It is encouraged that recyled material be used when practical as long as it meets the requirements of the specification (see 3.3).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-E-52246C(ME) Extractors, Track-Spike, Gasoline- or Diesel-Engine-Driven

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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